INCH-POUND
MS27474H
w/AMENDMENT 1
15 November 2012
SUPERSEDING
MS27474H
2 February 2012

DETAIL SPECIFICATION SHEET

# CONNECTORS, RECEPTACLE, ELECTRICAL, JAM NUT MOUNTING, CRIMP TYPE, BAYONET COUPLING, SERIES II

Reactivated after 24 May 2004 and may be used for new and existing designs and acquisitions.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-38999.

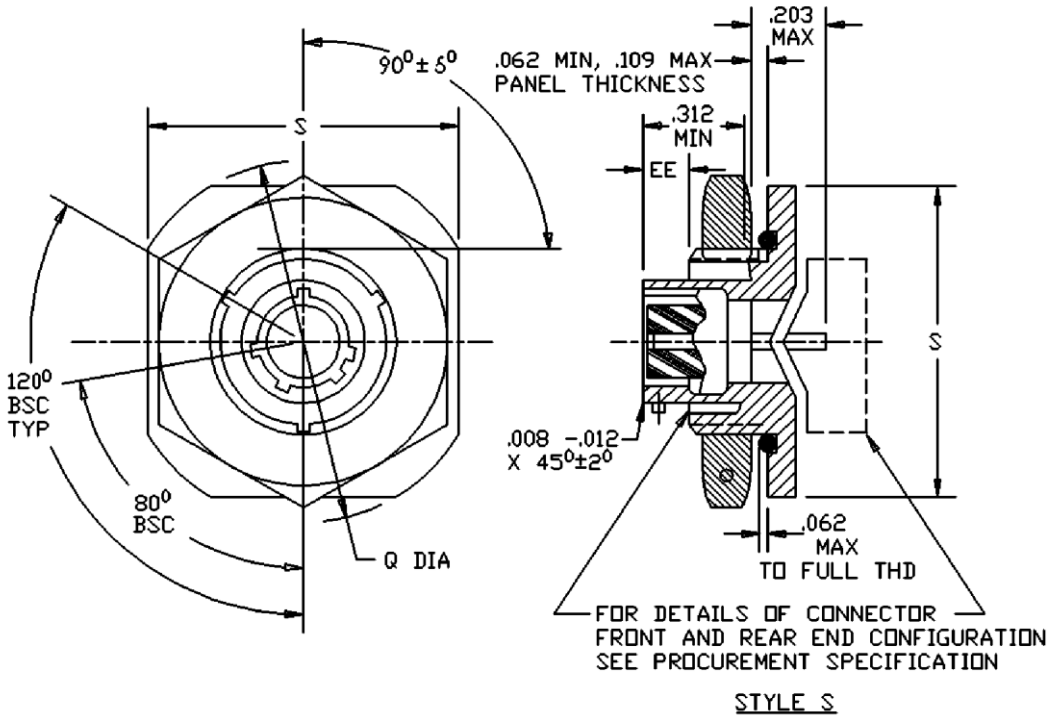
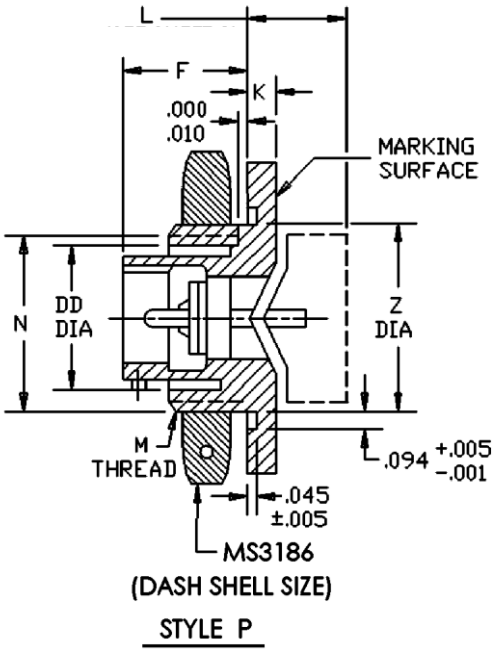



FIGURE 1. Receptacle, classes E, T and P.



Source: https://assist.dla.mil -- Downloaded: 2026-10-23T16:41Z
Check the source to verify that this is the current version before use.




| Shell size | DD dia +.010 -.000 | EE +.005 .000 | F ±.005 | K +.011 -.010 | M thd class 2A plated | N +.001 -.006 flat | Q dia ±.016 | S ±.016 | Z dia +.006 -.005 |
|---|---|---|---|---|---|---|---|---|---|
| 8 | .678 | .140 | .438 | .094 | .875 – 20 UNEF | .817 | 1.375 | 1.250 | .965 |
| 10 | .780 | | | | 1.000 – 20 UNEF | .941 | 1.500 | 1.375 | 1.090 |
| 12 | .963 | | | | 1.125 – 18 UNEF | 1.065 | 1.625 | 1.500 | 1.215 |
| 14 | 1.088 | | | | 1.250 – 18 UNEF | 1.190 | 1.750 | 1.625 | 1.340 |
| 16 | 1.222 | | | | 1.375 – 18 UNEF | 1.320 | 1.938 | 1.781 | 1.477 |
| 18 | 1.333 | | | | 1.500 – 18 UNEF | 1.440 | 2.016 | 1.890 | 1.602 |
| 20 | 1.458 | .166 | .464 | | 1.625 – 18 UNEF | 1.565 | 2.141 | 2.016 | 1.727 |
| 22 | 1.583 | | | | 1.750 – 18 UNS | 1.690 | 2.265 | 2.140 | 1.852 |
| 24 | 1.708 | | | | 1.875 – 16 UN | 1.815 | 2.390 | 2.265 | 1.977 |

FIGURE 1. Receptacle, classes E, T and P – Continued.



Source: https://assist.dla.mil -- Downloaded: 2026-10-23T16:41Z
Check the source to verify that this is the current version before use.

| Shell size | Nut MS3186 | L class E max length | L class T max length | L class P max length | Recommended packing (O-ring) (see note 7) |
|---|---|---|---|---|---|
| 8 | - 8 | .516 | .344 | .578 | AS3578 – 022 |
| 10 | - 10 | | | | AS3578 – 024 |
| 12 | - 12 | | | | AS3578 – 026 |
| 14 | - 14 | | | | AS3578 – 028 |
| 16 | - 16 | | | | AS3578 – 029 |
| 18 | - 18 | | | | AS3578 – 030 |
| 20 | - 20 | .484 | .312 | .547 | AS3578 – 031 |
| 22 | - 22 | | | | AS3578 – 032 |
| 24 | - 24 | | .375 | .516 | AS3578 – 034 |

FIGURE 1. Receptacle, classes E, T and P - Continued.



Source: https://assist.dla.mil -- Downloaded: 2026-10-23T16:41Z
Check the source to verify that this is the current version before use.

| Inches | mm | Inches | mm | Inches | mm | Inches | mm |
|---|---|---|---|---|---|---|---|
| .001 | 0.03 | .312 | 7.92 | 1.088 | 27.64 | 1.625 | 41.28 |
| .005 | 0.13 | .344 | 8.74 | 1.090 | 17.69 | 1.690 | 42.93 |
| .006 | 0.15 | .375 | 9.53 | 1.125 | 28.58 | 1.708 | 43.38 |
| .007 | 0.18 | .438 | 11.13 | 1.190 | 30.23 | 1.727 | 43.87 |
| .008 | 0.20 | .464 | 11.79 | 1.215 | 30.86 | 1.750 | 44.45 |
| .010 | 0.25 | .484 | 12.29 | 1.222 | 31.04 | 1.781 | 45.24 |
| .011 | 0.28 | .516 | 13.11 | 1.250 | 31.75 | 1.815 | 46.10 |
| .012 | 0.30 | .547 | 13.89 | 1.320 | 33.53 | 1.852 | 47.04 |
| .016 | 0.41 | .578 | 14.68 | 1.333 | 33.86 | 1.875 | 47.63 |
| .045 | 1.14 | .678 | 17.22 | 1.340 | 34.04 | 1.890 | 48.00 |
| .062 | 1.57 | .780 | 19.81 | 1.375 | 34.95 | 1.977 | 50.22 |
| .094 | 2.39 | .817 | 20.75 | 1.440 | 36.58 | 2.016 | 51.21 |
| .109 | 2.77 | .875 | 22.23 | 1.458 | 37.03 | 2.140 | 54.36 |
| .139 | 3.53 | .941 | 23.90 | 1.477 | 37.52 | 2.265 | 57.53 |
| .140 | 3.56 | .963 | 24.46 | 1.500 | 38.10 | 2.390 | 60.71 |
| .166 | 4.22 | .965 | 24.51 | 1.565 | 39.75 | | |
| .203 | 5.16 | 1.000 | 25.40 | 1.583 | 40.21 | | |
| .223 | 5.66 | 1.065 | 27.05 | 1.602 | 40.69 | | |

NOTES:

1. Dimensions are in inches.
2. Metric equivalents are given for information only.
3. Jam nut mounting receptacles shall be provided with an O-ring.
4. Sides of groove may have 5° maximum taper.
5. For main keyway polarization and keyway positions, see MIL-DTL-38999.
6. Diameters M, Q and Z shall be concentric within .015 inches (0.381 mm) T.I.R.
7. Recommended packing is for dimensions only, material is optional.
8. Applicable potting forms for class P: SAE-AS85049/58 and SAE-AS85049/75.

FIGURE 1. Receptacle, classes E, T and P - Continued.



Source: https://assist.dla.mil -- Downloaded: 2026-10-23T16:41Z
Check the source to verify that this is the current version before use.

REQUIREMENTS:

Dimensions and configurations: See figure 1.

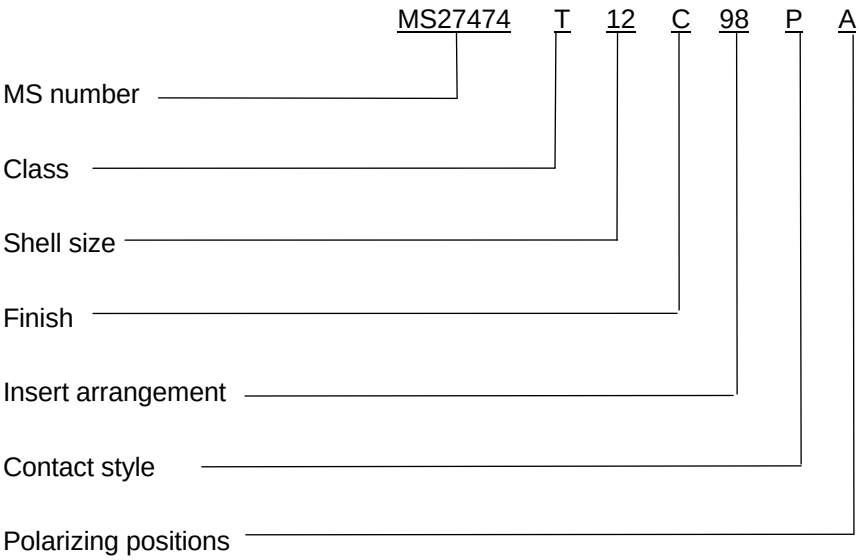
Interface dimensions shall be in accordance with MIL-DTL-38999.

This receptacle mates with MS27473, MS27480, MS27484 and MS27500.

Insert arrangements: See MIL-STD-1560.

For class E, compression nut in accordance with SAE-AS85049/27 shall be included.

Part or Identifying Number (PIN) example:




No letter is required for normal.

Restrictions:
Class E is inactive for new design.
Class P is inactive for new design for the Air Force.
Finish F is not for Navy use. Finish F is inactive for new design for the Air Force.

Amendment notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

Referenced documents. In addition to MIL-DTL-38999, this document references the following:

MIL-STD-1560
MS3186
MS27473
MS27480
MS27484
MS27500
SAE-AS85049/27
SAE-AS85049/58
SAE-AS85049/75



Source: https://assist.dla.mil -- Downloaded: 2026-10-23T16:41Z
Check the source to verify that this is the current version before use.

CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - AS
Air Force – 85
DLA - CC

Preparing activity:
DLA - CC

(Project 5935-2012-197)

Review activities:
Army - AR, MI
Navy - EC, MC, OS
Air Force - 19, 99
DLA - IS

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.



Source: https://assist.dla.mil -- Downloaded: 2026-10-23T16:41Z
Check the source to verify that this is the current version before use.